TrueJES: the jet energy scale along the truth ladder

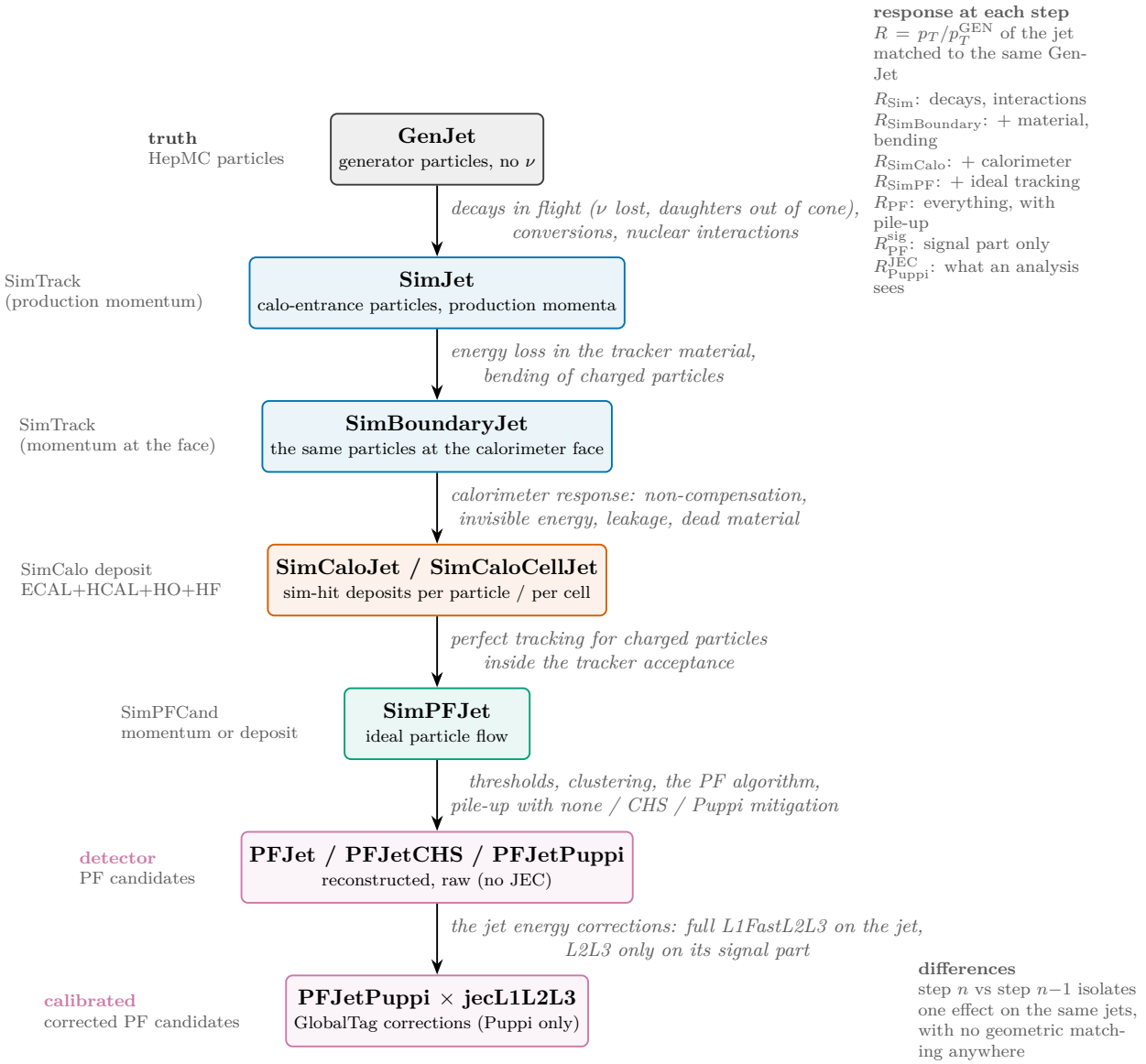
illustrations

Reco2Pico / analysis/TrueJES

October 18, 2026

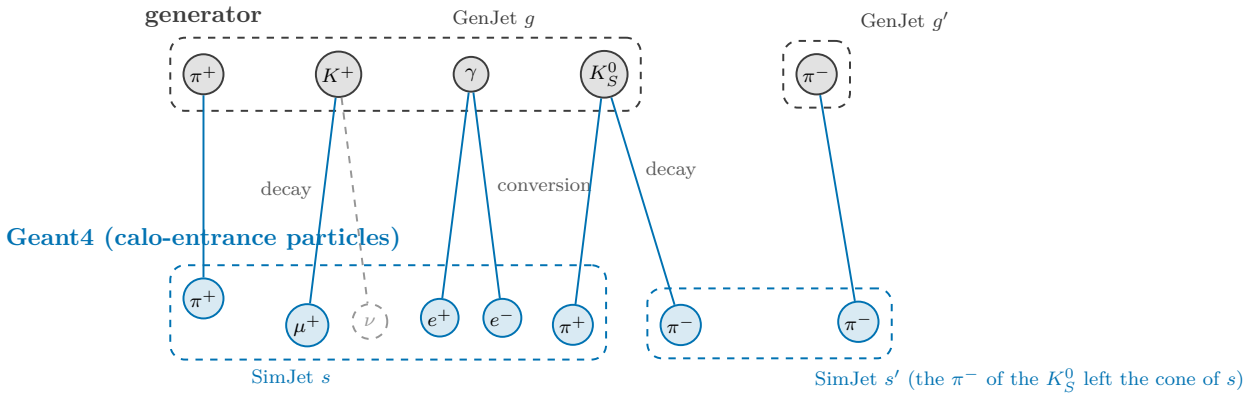
1. The ladder

One jet collection per step, all built on the same simulated events and all matched to the same generator jet. The difference of the response between two consecutive steps isolates one effect; the last rung is the Puppi jet with the GlobalTag corrections, i.e. what an analysis sees.



2. Matching by ancestry, not by ΔR

Every Geant4 particle descends from one generator primary, and every generator particle sits in at most one gen jet. The energy a jet shares with a gen jet is therefore exact, whatever the magnetic field and the decays did to the directions. The same scores are used at every step.



every SimTrack has one generator ancestor (walk `SimTrack_parentIdx` up to the primary) **every generator particle sits in at most one GenJet** (gen jets re-clustered in the job)

for a pair (s, g): $E_{\text{shared}} = \sum_{c \in s,\ \text{anc}(c) \in g} E_c = E_{\pi^+} + E_{\mu^+} + E_{e^+} + E_{e^-} + E_{\pi^+}$

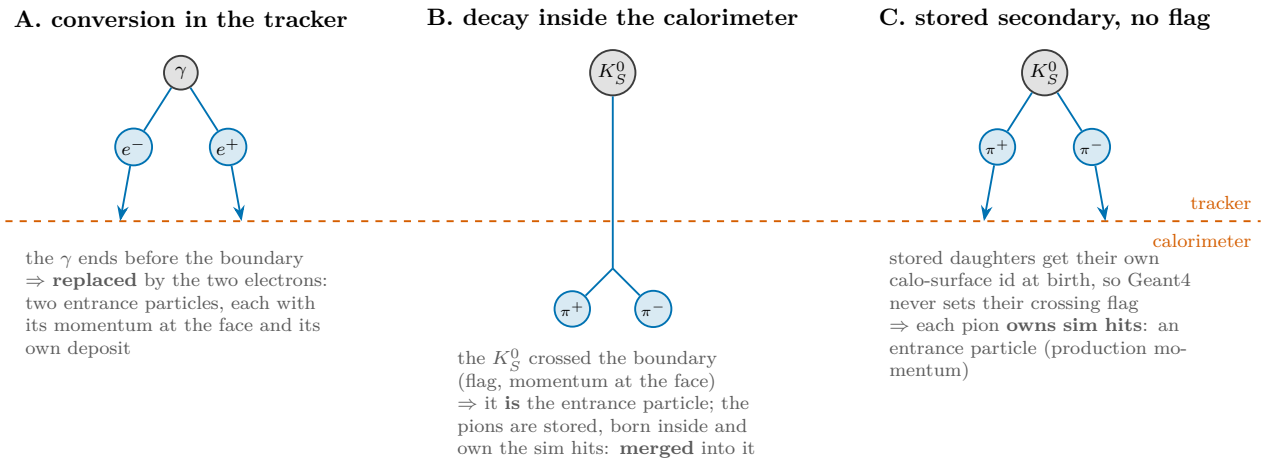
`simFraction` $= E_{\text{shared}}/E(s)$: purity of s `genFraction` $= \dfrac{1}{E(g)} \displaystyle\sum_{p \in g} E_p\, f_s(p)$: efficiency of g into s, with $f_s(K^0_S) = \dfrac{E_{\pi^+}}{E_{\pi^+} + E_{\pi^-}}$, $f_s(K^+) = 1$ (the ν is not clustered)

matched: the best pair for g with both fractions > 0.5; `ptRatio` $= p_T(s)/p_T(g)$ is the response of the pair

no ΔR anywhere: immune to the bending in the field and to the decay kinematics at low p_T

3. Which Geant4 particles are clustered

The calorimeter-entrance particles: the cut through the stored decay tree at the tracker/calorimeter boundary, after the decays, conversions, energy loss and bending in the tracker. Geant4's crossing flag alone does not give it (a stored secondary born in the tracker never carries it, and the sim hits of a daughter born inside the calorimeter reference the daughter), so the flag, the production vertex and the ownership of the sim hits are combined, and every hit is bookkept to exactly one entrance particle. The same list defines the sim jets (production momenta, and at the face), the deposits and the ideal particle-flow list.



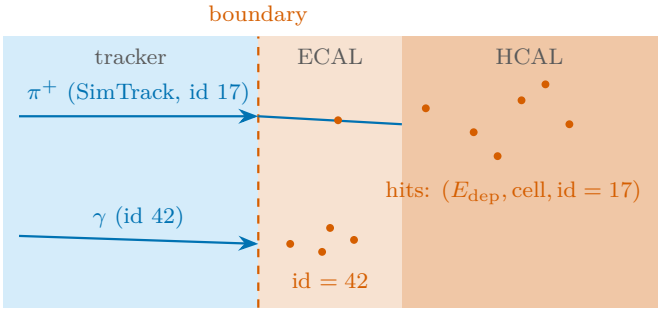
entrance particle ⇔ no entrance ancestor, and: crossed the boundary, or born in the tracker and reaching the calorimeter (owns sim hits, or daughters born outside the tracker do)

every sim hit is bookkept to exactly one entrance particle

(`SimTrack_isEntrance`, `SimTrack_entranceIdx`, `SimTrack_ownsCaloHits`)

4. The calorimeter response from the sim hits

The calorimeter sim hits carry the Geant4 id on the calorimeter surface, which is mapped to the entrance particle, so the deposits are summed per particle without clustering; jets of the deposits give the calorimeter-only response, jets of the cells the same energy with the calorimeter geometry, and an ideal particle flow built from momenta and deposits shows what tracking recovers.



SimCalo: one row per particle that entered `geantTrackId` of every PCaloHit = the particle at the calo surface (CaloSD)
⇒ **no clustering**: sum the hits per id

ECAL hits in GeV; HCAL hits relabelled and × digitiser sampling factors (GeV-equivalent); HF photoelectrons × HF factor (unvalidated)

cells summed over all particles first → optional per-cell threshold (rechit-like) → shared in proportion to the deposits

`response` = $E_{\text{deposit}}/E_{\text{at boundary}}$ (single particle)
`SimCaloJet` = anti-k_T 0.4 of the deposits (direction = energy-weighted cell direction)

SimPFCand: ideal particle flow

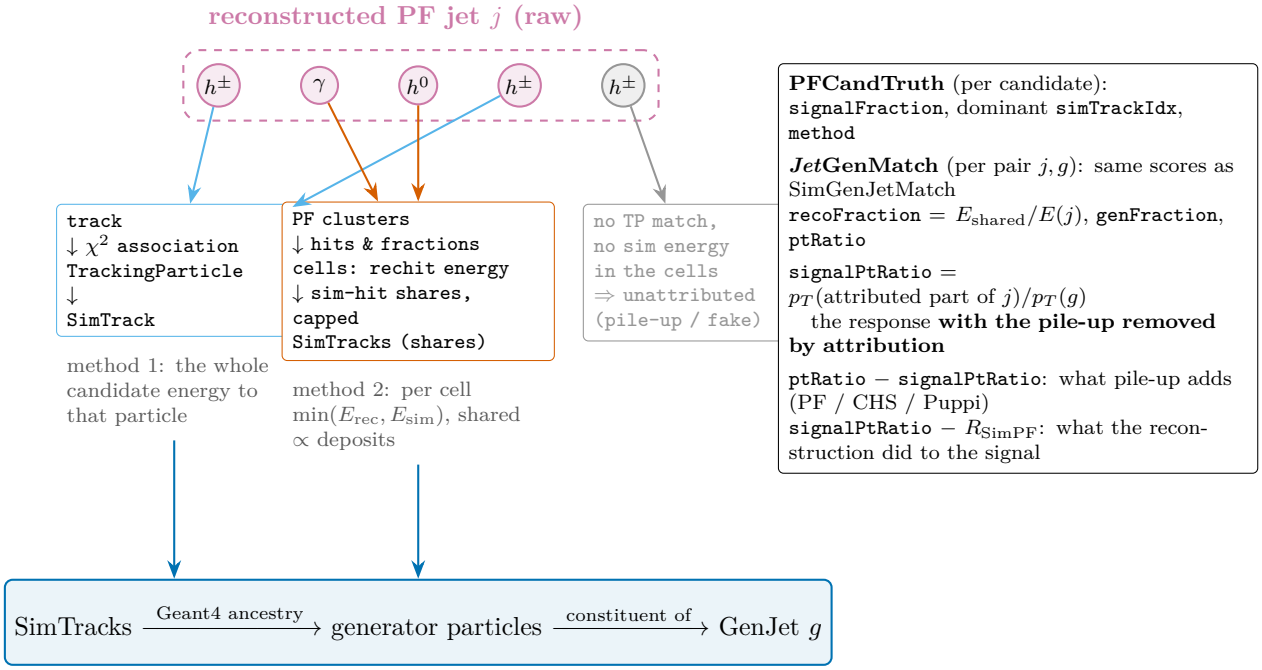
calo-entrance particle, charged, born inside the tracker volume, $|\eta| < 2.5$, $p_T > 0.5$ GeV
→ its **momentum** (perfect tracking), kind 1
otherwise → its **deposit** (neutral: kind 2, charged not trackable: kind 3), or nothing if it left none

`SimPFJet` = anti-k_T 0.4 of these; SimPF − SimCalo = what tracking recovers

5. The reconstructed jets on the same ladder

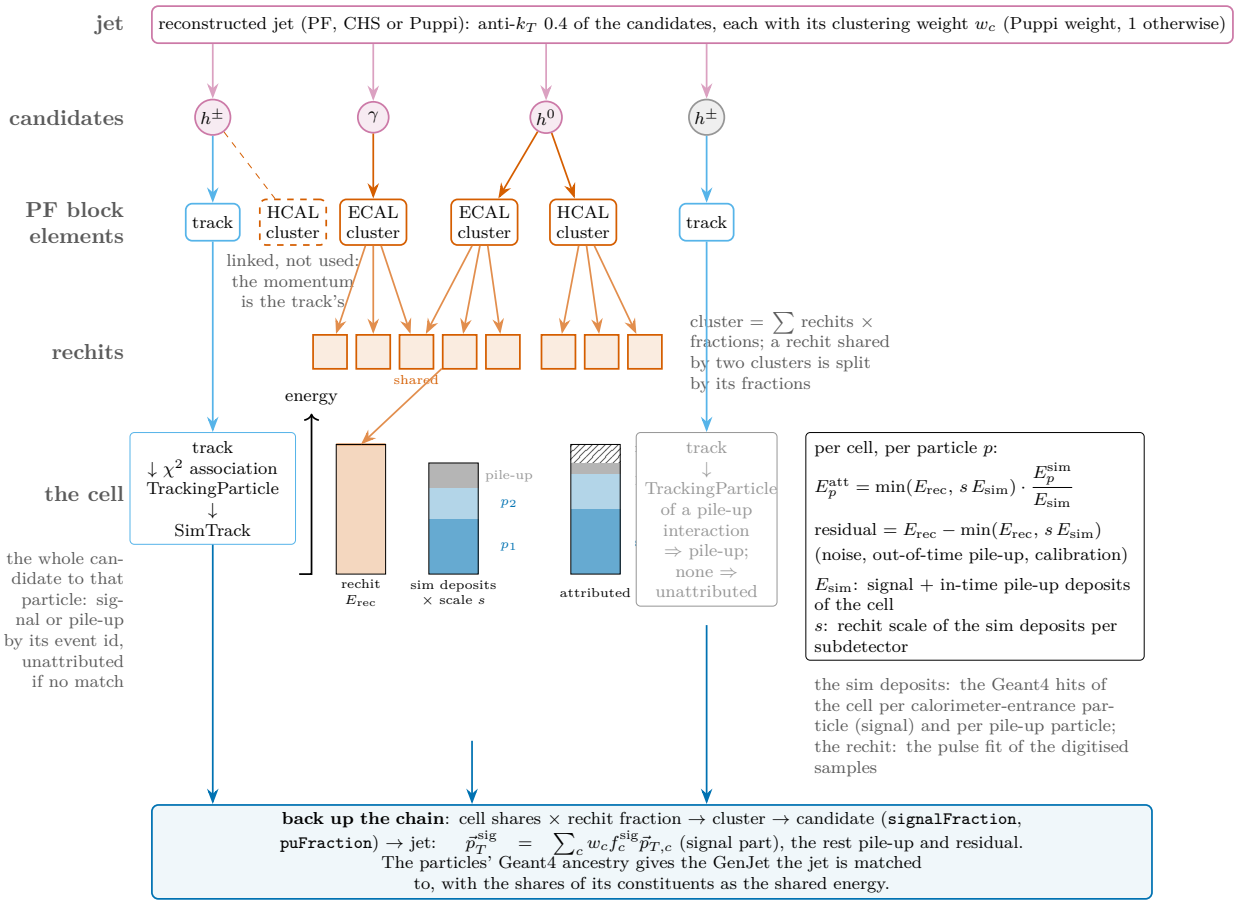
PF candidates are attributed to simulated particles (track association for charged ones, sim-hit shares of the cluster cells for the others); what is not attributed is pile-up, noise or fakes. The reconstructed

jets are then matched by the same ancestry scores, and the signal-only response separates the pile-up from the reconstruction effects without matching pile-up conditions.



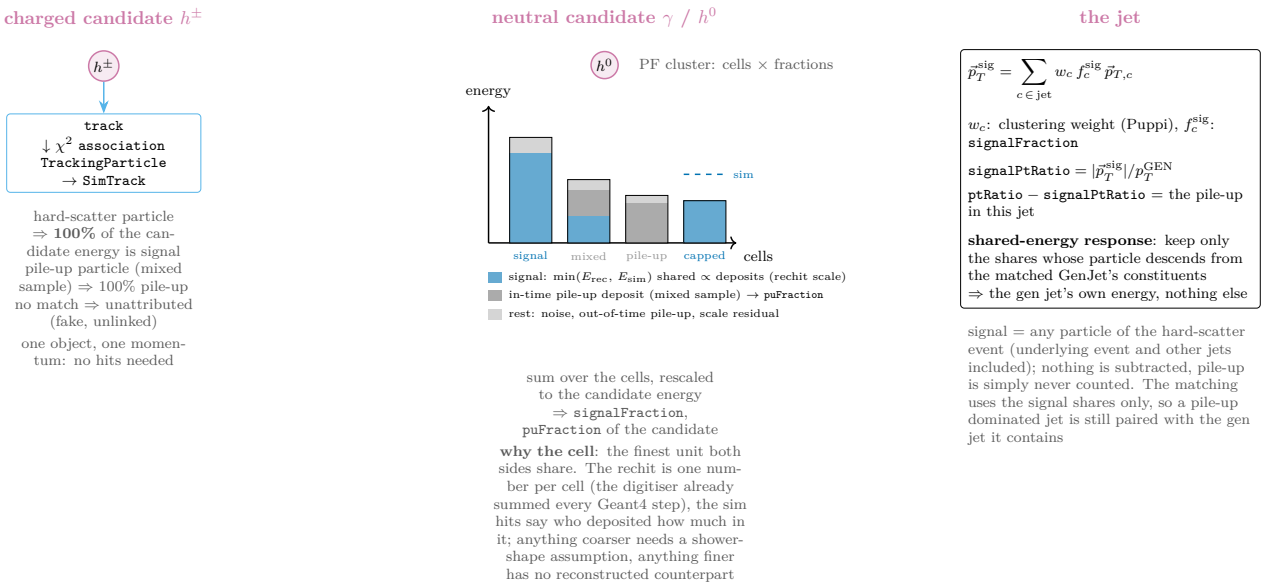
6. From the jet down to the cell

Everything the reconstruction side of the ladder uses is taken apart to the rechits: the jet to its candidates (with their clustering weights), each candidate to the block elements it was built from, a track for the charged ones and the PF clusters otherwise, each cluster to its rechits with their fractions. The rechit is one cell, and in that cell the Geant4 deposits are known per calorimeter-entrance particle of the signal and per pile-up particle. The attribution is made there, cell by cell, and summed back up the same chain to the candidate and the jet; the track side goes through the TrackingParticle instead. The rechit itself is the pulse fit of the digitised samples, so noise, out-of-time pile-up and the calibration stay in the residual.

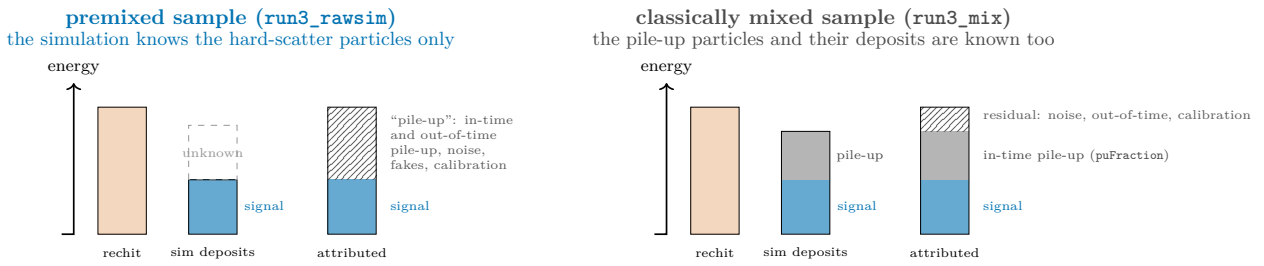


7. How the pile-up is removed: attribution, not subtraction

Every PF candidate is split into the part that can be traced to a simulated particle of the hard-scatter event and a remainder. Charged candidates are traced through their track, one object with one momentum. Neutral ones are traced cell by cell: the cell is the finest unit the reconstruction and the simulation share (the rechit is one number per cell, the sim hits say who deposited how much in it), anything coarser needs a shower-shape assumption, anything finer has no reconstructed counterpart. Within a cell the reconstructed energy, capped by the simulated one on the rechit scale, is shared in proportion to the deposits, which is the only unbiased statement left once the fluctuation of a cell cannot be assigned to a particle. The remainder is pile-up, noise or a fake. The signal part of a jet is the sum of the traced parts of its constituents, so no pile-up matching between samples is needed.



The same attribution runs on both kinds of sample. On the premixed one the simulation knows the hard-scatter particles only, so the cap is the signal deposit and everything unattributed is "pile-up" in one number: in-time and out-of-time pile-up, noise, fakes and the calibration residual of the cap. On the classically mixed one the pile-up particles and their deposits are known, the cap includes them, they get their own shares, and only noise, out-of-time pile-up and calibration remain in the residual. The signal part is defined identically, which is what makes the two sets of results comparable.



cell: cap = $\min(E_{\text{rec}}, s\,E_{\text{sim}}^{\text{signal}})$, shared among the signal particles ∝ deposits; the rest is unattributed

track: TrackingParticle of the hard-scatter event ⇒ signal; none ⇒ nothing (every pile-up track lands here)

candidate: signalFraction; 1 − signalFraction = pile-up + noise + fakes + calibration, in one number

jet: signalPtRatio; ptRatio − signalPtRatio = the unattributed part, split by candidate type only (section 4.6)

cell: cap = $\min(E_{\text{rec}}, s\,(E_{\text{sim}}^{\text{signal}} + E_{\text{sim}}^{\text{pile-up}}))$, shared among signal and pile-up particles ∝ deposits; the rest is the residual

track: TrackingParticle of the hard-scatter event ⇒ signal; of a pile-up interaction ⇒ pile-up; none ⇒ residual

candidate: signalFraction, puFraction, residual = 1− both

jet: the same signalPtRatio; the unattributed part split into true in-time pile-up by origin and residual; the pile-up rungs SimPFPUJet, SimPFPUCHSJet, SimCaloCellPUJet

the signal part is defined identically on both samples: same cap for the signal, same shares, same signalPtRatio and shared-energy response.
The ladders agree within a few percent; the mixed sample adds what the unattributed part is made of.

8. The pile-up truth: classical mixing

On the premixed samples the pile-up is a pre-digitised library whose particles are gone, so the remainder of the attribution is "pile-up, noise or fake" as a whole. Mixing the signal GEN-SIM with minimum-bias GEN-SIM events the classical way keeps every pile-up particle and deposit, tagged by interaction and crossing. Every cell is then signal plus known pile-up plus the rest, the calorimeter scales can be measured cell by cell, the reconstructed candidates get a pile-up share next to their signal share, and the simulation-level ladder gets jets with the pile-up in them: the ideal PF of signal

plus pile-up, and the same after a perfect charged-hadron subtraction. The pile-up of a jet is thereby measured three ways on the same events.

classical mixing: the pile-up truth

signal GEN-SIM
Geant4 particles and hits

↓

minimum-bias GEN-SIM
the pile-up library of the campaign

↓

DIGI with the MixingModule
the campaign's pile-up profile, in-time and out-of-time crossings

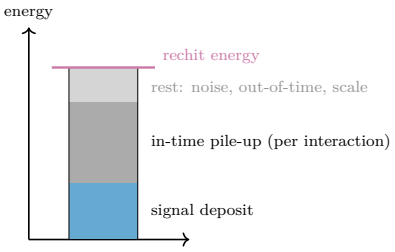
↓

picoMixedTruth, kept from the transient CrossingFrames:
in-time interactions: SimTracks, SimVertices, calorimeter sim hits, each tagged (interaction, Geant4 id)
out-of-time: deposits per (cell, crossing)
TrackingParticles of the pile-up too

↓

RECO → Pico
`pileupTruth=picoMixedTruth`

every calorimeter cell



the sim deposits are put on the rechit scale with factors measured here, cell by cell and per subdetector (the HF gain among them)

`SimCaloCell: energy, puEnergy, ootEnergy, nPU`

the jets with the pile-up in them (anti-k_T 0.4, matched to the gen jet through their signal constituents)

SimCaloCellPUJet: the cells with their signal + in-time pile-up deposits, the calorimeter truth with pile-up, no noise, no thresholds

SimPFPUJet: the ideal PF of the signal and the pile-up entrance particles (charged trackable from momentum, the rest from deposits), the counterpart of the raw PF jet

SimPFPUCHSJet: the same without the trackable charged pile-up, a perfect charged-hadron subtraction, the counterpart of the CHS jet: what remains is the neutral and untrackable pile-up

the pile-up of one jet, three ways

1. attribution (any sample)
`ptRatio` − `signalPtRatio`: the raw jet minus its signal part, candidate by candidate

2. truth, no reconstruction (mixed sample)
$R_{\text{SimPFPU}} - R_{\text{SimPF}}$: the ideal PF with and without the pile-up particles; $R_{\text{SimPFPUCHS}}$: after a perfect charged-hadron subtraction

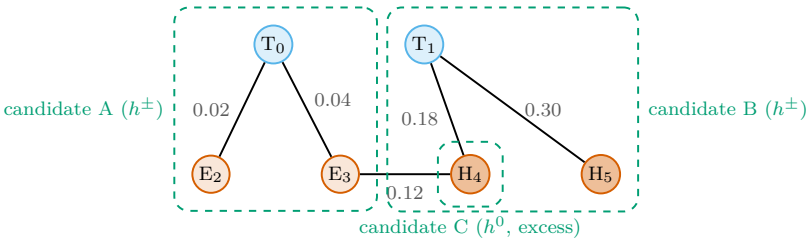
3. truth through the reconstruction (mixed sample)
`puFraction` of the constituents: the offset split into true in-time pile-up by candidate type and a residual (noise, out-of-time, scale)

2 − 1: what the reconstruction adds to the pile-up (thresholds, clustering, calibration)
1 − 3: the attribution convention SimPFPU vs PF and SimPFPUCHS vs CHS: the reconstruction and the vertex assignment at work; Puppi has no ideal of its own and is read against SimPF

9. The particle-flow block tables

For the reconstruction studies the Pico also keeps the full PF block structure: the elements, the geometric links PF found between them, and the elements each candidate finally used.

one PF block: elements and stored links



`PFBlockElem` - the inventory
one row per element: type, block, pointers to Track / PFCluster rows, candIdx

`PFBlockLink` - what PF found connected
one row per stored link: the two elements and the linking distance ($\Delta\eta$-$\Delta\phi$ at the calo entrance, placeholders for same-object links)
input of the algorithm, not its decision

`PFCandElem` - what PF decided
one row per (candidate, element) used
A: T_0, E_2, E_3 B: T_1, H_5 C: H_4

the physics is in the difference: H_4 was linked to both tracks but used by neither; T_0 was linked to E_3 and kept it

Where to read more

- `Reco2Pico/docs/TrueJES.md`: the physics note
- `Reco2Pico/PicoProducer/README.md`: the table definitions
- `Reco2Pico/analysis/TrueJES/README.md`: the analysis